A Fast Iterative Multipath Parameter Estimation Algorithm Based on the Maximum Likelihood Criterion for GNSS Receivers

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***Abstract*—In Global Navigation Satellite Systems (GNSS), multipath effects can cause severe measurement errors. To address this problem, this paper proposes a fast iterative optimization algorithm based on the maximum likelihood (ML) criterion to estimate the complex amplitudes and delay parameters of multipath signals. The proposed algorithm adopts a stepwise optimization strategy. First, a closed-form solution for the complex amplitude parameters is obtained using the ML method. Then, a regularized two-step Levenberg-Marquardt (RTS-LM) optimization algorithm is developed to iteratively estimate the delay parameters. RTS-LM applies the TS-LM algorithm to GNSS multipath scenarios and further improves TS-LM to ensure fast and robust convergence. Simulation results verify the effectiveness of the proposed algorithm, demonstrating its capability of delay super-resolution estimation as well as superior parameter estimation accuracy, convergence rate, and robustness against multipath.**

***Index Terms*—Multipath parameter estimation, maximum likelihood, stepwise optimization, approximate LM step**

I. Introduction

In complex environments such as urban and forested areas, multipath errors severely affect the measurement accuracy and reliability of Global Navigation Satellite Systems (GNSS) receivers [1]. Current multipath mitigation techniques can be categorized into antenna design, tracking loop design, and multipath parameter estimation. Although antenna design techniques (such as choke-ring antennas [2] and array antennas [3]) and tracking loop design techniques (such as narrow correlators [4] and high-resolution correlators [5]) can suppress multipath errors to some extent, they suffer from issues such as high implementation complexity and incomplete multipath mitigation. Multipath parameter estimation methods can be broadly classified into two categories: compressive sensing-based methods and maximum likelihood (ML) or least squares (LS)-based methods. Compressive sensing-based algorithms estimate multipath parameters by exploiting sparse reconstruction. Reference [6] employs the Orthogonal Matching Pursuit method to estimate delay parameters, Reference [7] obtains a sparse solution through the deconvolution extension of the sparse randomized Kaczmarz algorithm, and References [8] and [9] estimate multipath parameters by minimizing the l_1-norm. However, these algorithms rely on a discrete grid-based model, and when the true signal parameters are off the grid, large estimation errors may arise due to model mismatch. ML-based and certain LS-based techniques are grid-less algorithms and thus do not suffer from off-grid errors. Reference [10] proposes a Multipath Estimating Delay Lock Loop (MEDLL) that uses a ML estimator to perform delay estimation, but the algorithm exhibits poor robustness and noise-resilience performance. References [11] and [12] proposed optimization algorithms for delay estimation based on the ML criterion and the LS criterion, respectively. Reference [13] estimates multipath parameters using the Expectation-Maximization (EM) algorithm. However, both the aforementioned optimization algorithms and the EM algorithm suffer from slow convergence and high computational complexity, and their performance remains to be further improved.

Reference [14] employs the Levenberg-Marquardt (LM) method for GNSS direct signal tracking based on the ML criterion. LM is an optimization technique that combines the advantages of the steepest descent and Newton methods. Reference [15] proposes a Levenberg Marquardt-Quasi Newton (LM-QN) optimization algorithm for carrier tracking. Reference [16] proposes a Two-step Levenberg-Marquardt (TS-LM) algorithm, which approximates the Hessian matrix (HM) using $\mathbf{J}^T\mathbf{J}$ (where $\mathbf{J}$ is the Jacobian matrix) and adjusts the damping factor based on the residual norm. In each iteration, TS-LM computes not only the LM step but also an approximate LM step, thereby reducing the computational burden associated with evaluating the Jacobian matrix.

This paper proposes a fast iterative optimization algorithm based on the ML criterion for estimating the complex amplitudes and delay parameters of GNSS multipath signals. The proposed algorithm adopts a stepwise optimization strategy: the complex amplitudes are first solved in closed form using the ML method, followed by iterative delay estimation using a Regularized Two-step Levenberg-Marquardt (RTS-LM) optimization algorithm. It should be noted that the proposed ML method employs the Moore-Penrose pseudo-inverse instead of the conventional matrix inversion, avoiding issues caused by nearly singular matrices and improving the algorithm's robustness. The proposed RTS-LM improves TS-LM and applies it to GNSS multipath scenarios. By computing the full HM, the errors induced by the $\mathbf{J}^T\mathbf{J}$ approximation are avoided. Meanwhile, the damping parameter is updated based on the gain ratio of the cost function (CF). Furthermore, by

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computing both the LM step and the approximate LM step at each iteration, the computational cost associated with the HM is reduced while the convergence speed is improved. In addition, RTS-LM adaptively selects the current iteration step based on the iteration progress, avoiding oscillations or divergence in the error curve caused by excessively large steps, thereby ensuring stable convergence of the algorithm and further improving its robustness. The simulation results demonstrate that the proposed algorithm is effective and exhibits super-resolution capability in delay estimation, while achieving superior measurement accuracy, convergence speed, and multipath robustness compared with the benchmark algorithms.

II. Received Signal Modeling and Analysis

For GNSS multipath scenarios, it is assumed that the carrier and navigation data components in the received signal have been removed and that the Doppler frequency shift has been compensated. The sampled discrete complex baseband received signal can then be expressed as [17]:

$$r\left(kT_s\right)=\sum_{l=1}^{P}a_l\cdot c\left(kT_s-\tau_l\right)+n\left(kT_s\right);\quad k=0,1,\ldots,N-1 \tag{1}$$

where the path index $l=1,2,...,P$, a_l and τ_l denote the complex amplitude and delay of the l-th path signal, respectively. $c\left(kT_s\right)$ represents the C/A code, $n\left(kT_s\right)$ denotes complex white Gaussian noise, and T_s is the sampling interval. When the noise variance at the receiver is σ^2, the joint probability density function of the N received signal samples is given by:

$$p\left(\boldsymbol{\theta}|\mathbf{r}_N\right)=\frac{1}{\left(\pi\sigma^2\right)^N}\exp\left[-\frac{1}{\sigma^2}\left(\mathbf{r}_N-\mathbf{r}'_N\right)^H\left(\mathbf{r}_N-\mathbf{r}'_N\right)\right] \tag{2}$$

Here, the vector of unknown parameters is $\boldsymbol{\theta}=[\mathbf{a}^T,\boldsymbol{\tau}^T]^T$, where $\mathbf{a}=[a_1,a_2,\ldots,a_P]^T$ is the complex amplitude parameter vector, $\boldsymbol{\tau}=[\tau_1,\tau_2,\ldots,\tau_P]^T$ is the delay parameter vector, $\mathbf{r}_N=\left[r\left(0\right),r\left(T_s\right),\ldots,r\left(\left(N-1\right)T_s\right)\right]^T$ is the received signal sample vector, and $\mathbf{r}'_N$ is the locally reconstructed signal sample vector. Meanwhile, (2) can also be regarded as the likelihood function of $\boldsymbol{\theta}$. In this paper, the CF is defined as the negative log-likelihood function (ignoring $\frac{1}{\sigma^2}$); by substituting (1) into (2), the expression of the CF is derived as follows:

$$L(\boldsymbol{\theta}|\mathbf{r}_N)=\sum_{k=0}^{N-1}\left|r(kT_s)-\sum_{l=1}^{P}a_lc(kT_s-\tau_l)\right|^2 \tag{3}$$

III. Proposed Parameter Estimation Method

This section proposes an optimization algorithm based on the ML criterion, in which the complex amplitudes and delay parameters are optimized in a stepwise manner. In each iteration, the complex amplitudes are first estimated using the ML method, followed by delay estimation via RTS-LM. As the iterations proceed, the estimation errors of the proposed algorithm are rapidly and accurately reduced until convergence.

A. Preprocessing

Assuming that the number of paths P is known, the received signal is processed using a acquisition algorithm [18], yielding the initial delay values $\boldsymbol{\tau}^0$. The superscript of each variable denotes the current iteration. Subsequently, to improve the accuracy of the direct signal's initial value, a local correlation peak search is performed centered at τ_1^0, and the delay corresponding to the peak is taken as a more accurate initial value of the direct signal delay.

B. Regularized Two-step Levenberg-Marquardt Algorithm

After obtaining the delay estimate $\boldsymbol{\tau}^i$ in the i-th iteration, the complex amplitude $\mathbf{a}^{i+1}$ is obtained by solving $\frac{\partial L(\boldsymbol{\theta}|\mathbf{r}_N)}{\partial a_l}=0$ $(l=1,2,...,P)$. Note that the solution of $\mathbf{a}^{i+1}$ involves matrix inversion. To improve the robustness of the algorithm, the Moore-Penrose pseudo-inverse is used instead. The parameter vector after updating $\mathbf{a}^i$ is defined as $\boldsymbol{\eta}^{i+1}=[(\mathbf{a}^{i+1})^T,(\boldsymbol{\tau}^i)^T]^T$.

In this subsection, RTS-LM is proposed to iteratively estimate the delay parameters. To reduce computational complexity and accelerate convergence, in each iteration, RTS-LM computes not only the LM step but also an approximate LM step. The LM step of the delay parameter vector at the $(i+1)$-th iteration is expressed as:

$$\mathbf{d}_{\boldsymbol{\tau}}^{i+1}=-(\mathbf{H}(\boldsymbol{\eta}^{i+1})+\mu^i\mathbf{I})^{-1}\mathbf{g}(\boldsymbol{\eta}^{i+1}) \tag{4}$$

where μ^i is the damping coefficient, $\mathbf{I}$ is the identity matrix, and $\mathbf{g}(\boldsymbol{\eta}^{i+1})$ and $\mathbf{H}(\boldsymbol{\eta}^{i+1})$ denote the gradient and HM based on $\boldsymbol{\eta}^{i+1}$, respectively. Define $\mathbf{g}(\boldsymbol{\eta}^{i+1})\in\mathbb{R}^P$, $\left[\mathbf{g}(\boldsymbol{\eta}^{i+1})\right]_l=\frac{\partial L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l}$. Let the path index $h=1,2,...,P$ and $\mathbf{H}(\boldsymbol{\eta}^{i+1})\in\mathbb{R}^{P\times P}$ be defined such that when $l=h$, $\left[\mathbf{H}(\boldsymbol{\eta}^{i+1})\right]_{lh}=\frac{\partial^2 L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l^2}$; otherwise, $\left[\mathbf{H}(\boldsymbol{\eta}^{i+1})\right]_{lh}=\frac{\partial^2 L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l\partial\tau_h}$. Based on (3), the first- and second-order partial derivatives of the CF with respect to the delay are derived, and the expressions for $\frac{\partial L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l}$, $\frac{\partial^2 L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l^2}$, and $\frac{\partial^2 L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l\partial\tau_h}$ are given in (5)-(7), respectively.

$$\underset{(l\neq h)}{\frac{\partial^2 L(\boldsymbol{\eta}^{i+1})}{\partial\tau_l\partial\tau_h}}=2\sum_{k=0}^{N-1}\Re\left\{a_l^*a_h\cdot c'(kT_s-\tau_l)c'(kT_s-\tau_h)\right\} \tag{7}$$

where $c'(kT_s-\tau_l)$ and $c''(kT_s-\tau_l)$ are computed using the first- and second-order central difference formulas, respectively [11].

After obtaining the LM step $\mathbf{d}_{\boldsymbol{\tau}}^{i+1}$, it is used to compute the approximate LM step $\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1}$. Specifically, $\hat{\boldsymbol{\tau}}^{i+1}=\boldsymbol{\tau}^i+\mathbf{d}_{\boldsymbol{\tau}}^{i+1}$ is first defined, and the corresponding complex amplitude $\hat{\mathbf{a}}^{i+1}$ is then calculated, resulting in the parameter vector $\hat{\boldsymbol{\eta}}^{i+1}=[(\hat{\mathbf{a}}^{i+1})^T,(\hat{\boldsymbol{\tau}}^{i+1})^T]^T$. Subsequently, $\hat{\boldsymbol{\eta}}^{i+1}$ is substituted into (5) to obtain $\mathbf{g}(\hat{\boldsymbol{\eta}}^{i+1})$, which, together with $(\mathbf{H}(\boldsymbol{\eta}^{i+1})+\mu^i\mathbf{I})^{-1}$ in (4), is used to compute $\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1}$:

$$\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1}=-(\mathbf{H}(\boldsymbol{\eta}^{i+1})+\mu^i\mathbf{I})^{-1}\mathbf{g}(\hat{\boldsymbol{\eta}}^{i+1}) \tag{8}$$

The computation of $\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1}$ does not require evaluating $\mathbf{H}(\hat{\boldsymbol{\eta}}^{i+1})$, thereby reducing the computational load associated with the

$$\frac{\partial L(\boldsymbol{\eta}^{i+1})}{\partial \tau_l} = -2\sum_{k=0}^{N-1}\Re\left\{a_l^* c'(kT_s-\tau_l)\left[r(kT_s)-\sum_{\substack{h=1\\h\neq l}}^{P} a_h c(kT_s-\tau_h)\right]\right\} \tag{5}$$

$$\frac{\partial^2 L(\boldsymbol{\eta}^{i+1})}{\partial \tau_l^2} = -2\sum_{k=0}^{N-1}\Re\left\{a_l^* c''(kT_s-\tau_l)\left[r(kT_s)-\sum_{\substack{h=1\\h\neq l}}^{P} a_h c(kT_s-\tau_h)\right]\right\} \tag{6}$$

HM. In TS-LM, the iteration step is denoted as $\mathbf{s}_{\boldsymbol{\tau}}^{i+1} = \mathbf{d}_{\boldsymbol{\tau}}^{i+1} + \hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1}$. We define $\tilde{\boldsymbol{\tau}}^{i+1} = \boldsymbol{\tau}^i + \mathbf{s}_{\boldsymbol{\tau}}^{i+1}$, and the corresponding parameter vector is denoted as $\tilde{\boldsymbol{\eta}}^{i+1} = [(\mathbf{a}^{i+1})^T, (\tilde{\boldsymbol{\tau}}^{i+1})^T]^T$. With respect to $\boldsymbol{\tau}^i$, the actual and predicted reductions of the CF corresponding to $\tilde{\boldsymbol{\tau}}^{i+1}$ are given by:

$$\Delta L_{Are} = L(\boldsymbol{\eta}^{i+1}) - L(\tilde{\boldsymbol{\eta}}^{i+1}) \tag{9}$$

$$\Delta L_{Pre} = L(\boldsymbol{\eta}^{i+1}) - F(\mathbf{s}_{\boldsymbol{\tau}}^{i+1}) \tag{10}$$

where $F(\mathbf{s}_{\tau}^{i+1})$ is an approximate function of $L(\tilde{\boldsymbol{\eta}}^{i+1})$, defined as:

$$\begin{aligned} F(\mathbf{s}_{\boldsymbol{\tau}}^{i+1}) &= L(\boldsymbol{\eta}^{i+1}) + (\mathbf{d}_{\boldsymbol{\tau}}^{i+1})^T\mathbf{g}(\boldsymbol{\eta}^{i+1}) + (\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1})^T\mathbf{g}(\hat{\boldsymbol{\eta}}^{i+1}) \\ &+ \frac{1}{2}(\mathbf{d}_{\boldsymbol{\tau}}^{i+1})^T\mathbf{H}(\boldsymbol{\eta}^{i+1})\mathbf{d}_{\boldsymbol{\tau}}^{i+1} + \frac{1}{2}(\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1})^T\mathbf{H}(\boldsymbol{\eta}^{i+1})\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1} \end{aligned} \tag{11}$$

The gain ratio is defined as $\rho = \frac{\Delta L_{Are}}{\Delta L_{Pre}}$, and the damping coefficient μ^i is updated according to the trust-region method: when $\rho > 0$, $\mu^{i+1} = \mu^i \cdot \max\{\frac{1}{3}, 1-(2\rho-1)^3\}; v^{i+1} = 2$; otherwise, $\mu^{i+1} = \mu^i \cdot v^i; v^{i+1} = v^i \cdot 2$.

It is worth noting that, due to the large step size of TS-LM, oscillation or divergence may occur in the delay error curve during the iterative process, resulting in slow convergence or even failure to converge. To avoid these issues, RTS-LM selects the iteration step $\Delta\boldsymbol{\tau}^{i+1}$ according to the value of ρ: when $\rho > \frac{1}{4}$, $\Delta\boldsymbol{\tau}^{i+1} = \mathbf{s}_{\boldsymbol{\tau}}^{i+1}$; otherwise, $\Delta\boldsymbol{\tau}^{i+1} = \mathbf{d}_{\boldsymbol{\tau}}^{i+1}$. Finally, the delay parameter vector is updated for the current iteration as $\boldsymbol{\tau}^{i+1} = \boldsymbol{\tau}^i + \Delta\boldsymbol{\tau}^{i+1}$, and the updated vector of unknown parameters is given by $\boldsymbol{\theta}^{i+1} = [(\mathbf{a}^{i+1})^T, (\boldsymbol{\tau}^{i+1})^T]^T$.

The proposed algorithm is summarized in Table I.

TABLE I: Processing Procedure of the Proposed Algorithm

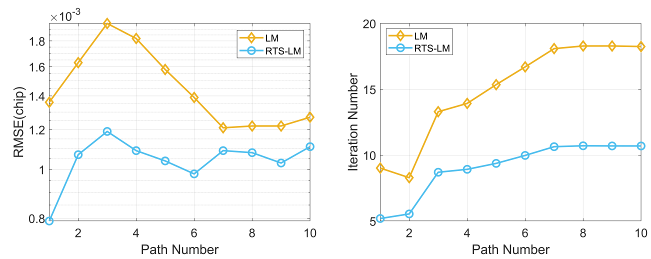
Step	Operations
Input:	$\mathbf{r}_N$, $\left\|\mathbf{g}(\boldsymbol{\theta}^{i+1})\right\|_\infty$ threshold ε_1, $\left\|\Delta\boldsymbol{\tau}^{i+1}\right\|_2$ threshold ε_2, maximum number of iterations U, and initial delay values.
Output:	estimation result of the vector of unknown parameters $\boldsymbol{\theta}$.
1	Initialization: set the iteration counter $i = 0$, and $v^0 = 2$.
2	Compute $\mathbf{a}^{i+1}$, and then evaluate $\mathbf{g}(\boldsymbol{\eta}^{i+1})$ and $\mathbf{H}(\boldsymbol{\eta}^{i+1})$ according to (5)-(7). If $i = 0$, set $\mu^1 = 10^{-6} \cdot \max\{\mathbf{H}(\boldsymbol{\eta}^1)\}$.
3	Compute the LM step $\mathbf{d}_{\boldsymbol{\tau}}^{i+1}$ according to (4), and use it to calculate the approximate LM step $\hat{\mathbf{d}}_{\boldsymbol{\tau}}^{i+1}$.
4	Compute the TS-LM step $\mathbf{s}_{\boldsymbol{\tau}}^{i+1}$, evaluate the gain ratio ρ according to (9)-(11), and update μ^i.
5	Select the RTS-LM iteration step $\Delta\boldsymbol{\tau}^{i+1}$ based on ρ, and set $\boldsymbol{\tau}^{i+1} = \boldsymbol{\tau}^i + \Delta\boldsymbol{\tau}^{i+1}$. $i = i+1$.
6	Stop the iteration if following conditions are satisfied: • $\left\|\mathbf{g}(\boldsymbol{\theta}^{i+1})\right\|_\infty < \varepsilon_1$ and $\left\|\Delta\boldsymbol{\tau}^{i+1}\right\|_2 < \varepsilon_2$. • $i \geq U$. Otherwise, go back to step 2.

IV. Simulation Studies

In this section, simulation experiments under different scenarios are conducted to validate the effectiveness of the proposed RTS-LM. Its measurement accuracy and convergence speed are evaluated and compared with those of the LM, LM-QN, and MEDLL algorithms. The root mean square error (RMSE) and the iteration numbers required for convergence are adopted as evaluation metrics for the measurement accuracy and convergence speed of the algorithms. The input signal-to-noise ratio (SNR) $\mathrm{SNR} = 10\log_{10}(A_1^2/\sigma^2)$, signal multipath ratio $\mathrm{SMR}_l = 10\log_{10}(A_1^2/A_l^2)$, multipath delay $\Delta\tau_l = \tau_l - \tau_1$, and multipath phase difference $\Delta\varphi_l = \varphi_l - \varphi_1$ are defined. The simulation system parameters are set as follows: a sampling rate of 10 MHz, a pseudo-code period, rate, and early/late spacing of 1 ms, 1.023 MHz, and 0.1 chip, respectively, and a coherent integration time of 1 ms. In the simulations, the maximum number of iterations for the algorithms is set to 150, and the delay RMSE and iteration count results of all algorithms are averaged over 1000 independent Monte Carlo simulation runs.

A. High-SNR Complex Multipath Scenario

In this subsection, the effectiveness of the proposed algorithm is validated in a complex multipath scenario with high-SNR, and its runtime performance is evaluated. The SNR is set to 10 dB, and the number of paths P in this scenario is varied sequentially from 1 to 10, with the corresponding path index set to $l = 1 \sim P$. The parameters of the direct signal are set to $\tau_1 = 306.9$ chip and $\varphi_1 = 0$ rad, while the parameters of the remaining multipath signals are set to $\mathbf{SMR} = [2, 2, 3.15, 4.6, 5.8, 6.1, 6.5, 7, 7.4]\,\mathrm{dB}$, $\Delta\boldsymbol{\tau} = [0.25, 0.88, 1.92, 2.55, 3.61, 4.82, 6.06, 6.88, 8.13]\,\mathrm{chip}$, and $\Delta\boldsymbol{\varphi} = \left[\pi/5, 2\pi/3, -4\pi/5, -\pi/2, \pi/5, \pi/2, \pi/4, -2\pi/3, -\pi/4\right]\mathrm{rad}$, respectively. Fig. 1(a)(b) respectively illustrate the direct signal delay measurement accuracy and the number of iterations for each algorithm under different numbers of paths. As shown in Fig. 1(a), the direct signal delay RMSE of RTS-LM is consistently lower than that of LM. When $P \geq 2$, the



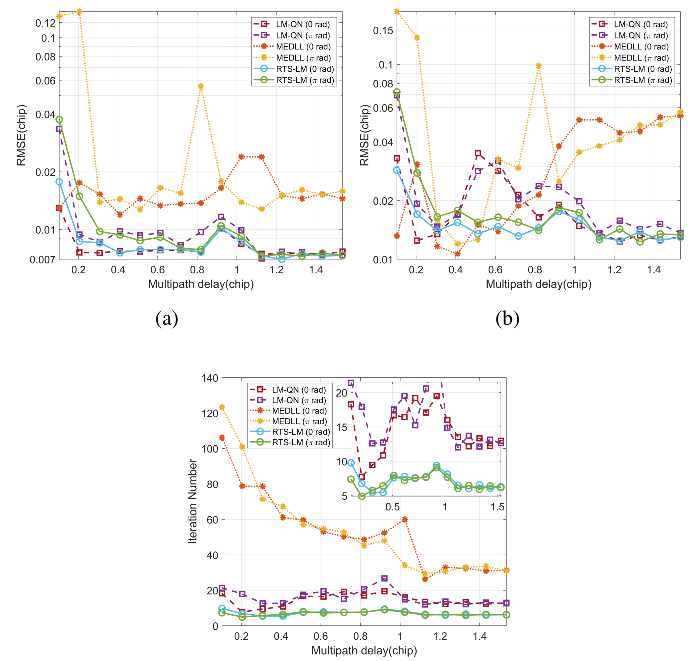
(a) (b)

Fig. 1: Direct signal delay RMSE and iteration numbers for different path numbers. (a) Delay. (b) Iteration number.

delay RMSE of RTS-LM exhibits only slight fluctuations within the range of 0.001 to 0.0012 chip, whereas the RMSE of LM shows much larger variations. As shown in Fig. 1(b), the maximum and minimum iteration counts of RTS-LM are reduced by approximately 7.6 and 3.8 iterations, respectively, compared with LM. In addition, the runtimes of RTS-LM and LM for completing ten parameter estimations are approximately 5.83 s and 7.55 s, respectively. In summary, RTS-LM can accurately estimate the direct signal delays in complex multipath scenarios, which validates the effectiveness of the proposed algorithm. Moreover, its measurement accuracy, convergence speed, runtime, and robustness to the number of multipaths are superior to those of LM.

B. Low-SNR Scenarios with Different Multipath Delays

In this subsection, the delay estimation performance of the proposed algorithm is evaluated in a low SNR two-path scenario. The simulation parameters are set as follows: the SNR is set to -10 dB, the direct signal parameters are $\tau_1 = 306.9$ chip and $\varphi_1 = 0$ rad, and the multipath signal parameters are $\mathrm{SMR}_2 = 5$ dB, $\Delta\tau_2 \in [0.1023, 1.5345]$ chip, and $\Delta\varphi_2 \in \{0, \pi\}$ rad. Fig. 2(a)-(c) respectively depict the delay measurement accuracy and iteration numbers of each algorithm for different multipath delay $\Delta\tau_2$ and phase $\Delta\varphi_2$. As shown in Fig. 2(a)(b), for most values of $\Delta\tau_2$, the direct and multipath signal delay RMSEs of MEDLL are significantly higher than those of the other algorithms, and exhibit relatively large fluctuations. For the direct signal, when $\Delta\tau_2 \leq 0.3069$ chip, the delay RMSE of LM-QN is slightly smaller than that of RTS-LM, while the performance of the two algorithms is comparable at the remaining time instants. For the multipath signals, the delay RMSE of LM-QN is larger than that of RTS-LM for most values of $\Delta\tau_2$, and exhibits significant fluctuations. When the multipath delay is below the delay resolution of GNSS signals (i.e., $\Delta\tau_2 < 0.5$ chip), the proposed RTS-LM algorithm is still capable of achieving super-resolution estimation of delay parameters with a low RMSE. As shown in Fig. 2(c), the number of iterations required by MEDLL is significantly larger than that of the other algorithms. Compared with LM-QN, the maximum and minimum numbers of iterations required by RTS-LM are reduced by approximately 16.9 and 2.9, respectively, and the iteration count is less sensitive to variations in $\Delta\tau_2$. In summary, RTS-LM not only achieves super-resolution estimation of delay in short-delay multipath scenarios ($\Delta\tau_2 < 0.5$ chip), but also outperforms the benchmark algorithms in terms of estimation performance, robustness to multipath delays, and convergence speed in medium- and long-delay multipath scenarios.



(a) (b)

(c)

Fig. 2: Delay RMSE and iteration number versus multipath delay and phase. (a) 1st path. (b) 2nd path. (c) Iteration number.

V. CONCLUSION

This paper proposes a fast iterative optimization algorithm based on the ML criterion, which efficiently and accurately estimates the complex amplitudes and delay parameters of multipath signals through a stepwise optimization strategy. After obtaining the initial delay values from the preprocessing output, the proposed algorithm first obtains a closed-form solution for the complex amplitude parameters using the ML method at each iteration. The delay parameters are then iteratively estimated via RTS-LM. This procedure is repeated until the stopping criterion is satisfied. Based on the computation of the LM step, RTS-LM further computes an approximate LM step, employs the full HM, and introduces an adaptive iteration-step selection mechanism. These strategies jointly improve the convergence speed and robustness of the algorithm. Simulation results demonstrate that the proposed algorithm is effective under different scenarios and achieves delay super-resolution, while exhibiting superior parameter estimation accuracy, convergence speed, and robustness to multipath compared with the benchmark algorithms.

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